INTELLIGENCE

Definition, Concept and Theories of Intelligence

Based on Morgan & King and Standard Academic Psychology Texts

✓ Definition & Concept	✓ Spearman's Two-Factor Theory	✓ Gardner's Multiple Intelligences	✓ Sternberg's Triarchic Theory

1. Intelligence: Definition and Concept

Intelligence is one of the most widely studied yet complex concepts in psychology. Psychologists differ in their definitions because intelligence involves several mental abilities such as reasoning, learning, problem-solving, adaptation, and abstract thinking.

Definitions of Intelligence

1. Wechsler's Definition

Intelligence is:

> *"The aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with the environment."*

This definition emphasizes three major aspects:

- Purposeful behavior
- Rational thinking
- Effective adaptation to surroundings

2. Morgan and King's View

Morgan and King describe intelligence as a general mental ability that enables individuals to:

- Learn from experience;
- Solve problems;
- Use knowledge effectively;
- Adapt to changing environments.

> *According to Morgan and King, intelligence is not merely the accumulation of facts but the ability to use acquired knowledge efficiently.*

Characteristics of Intelligence

#	Characteristic	Description
1	**Ability to Learn**	Intelligent individuals acquire new knowledge and skills more rapidly.
2	**Capacity for Abstract Thinking**	Intelligence allows understanding of concepts, symbols, and ideas beyond concrete experiences.

3	**Problem-Solving Ability**	It enables an individual to analyze situations and discover effective solutions.
4	**Adaptability**	A highly intelligent person adjusts successfully to new situations and environmental demands.
5	**Goal-Directed Behavior**	Intelligence helps individuals plan actions and work toward achieving objectives.
6	**Efficient Use of Experience**	Past experiences are utilized effectively in dealing with present and future problems.

2. Theories of Intelligence

Morgan and King discuss several theories explaining the structure and nature of intelligence. Among them, three important theories are:

- **Spearman's Two-Factor Theory**
- **Gardner's Multiple Intelligence Theory**
- **Sternberg's Triarchic Theory**

I Spearman's Two-Factor Theory

Charles Spearman, 1904

Spearman used factor analysis, a statistical method, to study intelligence. He observed that individuals performing well in one mental task often performed well in others. This led him to conclude that intelligence consists of two factors.

Components of the Theory

A. General Factor (g)	B. Specific Factor (s)
The **g-factor** refers to a general mental energy present in all intellectual activities. **Characteristics:** • Common to all intellectual tasks. • Responsible for overall intellectual efficiency. • Determines success in many areas. **Examples:** A student with high g-factor may perform well in Mathematics, Languages, Science, Logical reasoning.	Specific factors are abilities unique to particular tasks. **Characteristics:** • Different for different activities. • Responsible for specialized skills. • Operate along with the g-factor. **Examples:** Musical ability, Mechanical skill, Artistic talent, Verbal fluency.

Formula Representation

Intelligence = g + s

Every task requires some amount of general intelligence (g) and some amount of specific ability (s).

Example: Solving a mathematics problem requires General reasoning ability (g) + Mathematical skill (s).

Merits
- Introduced scientific measurement of intelligence.
- Supported by factor analytic studies.
- Explains positive correlations among mental abilities.

Criticisms
- Oversimplifies intelligence.
- Ignores social and creative abilities.
- Cannot explain exceptional talents adequately.

II Gardner's Multiple Intelligence Theory

Howard Gardner, 1983

Gardner argued that intelligence is not a single entity. Instead, humans possess several relatively independent intelligences.

Major Principles

- Every person possesses multiple intelligences.
- Individuals differ in the strength of each intelligence.
- Traditional IQ tests assess only limited abilities.
- Education should nurture all types of intelligence.

Types of Multiple Intelligences

#	Intelligence	Description	Examples	Key Characteristics
1	**Linguistic**	Ability to use language effectively.	Writers, Journalists, Lawyers	Good vocabulary; Storytelling skills; Reading and writing proficiency
2	**Logical-Mathematical**	Ability in reasoning and numerical operations.	Scientists, Mathematicians, Engineers	Problem-solving; Pattern recognition; Abstract thinking
3	**Spatial**	Ability to perceive and manipulate visual information.	Architects, Designers, Pilots	Mental imagery; Map reading; Visualization skills
4	**Musical**	Sensitivity to rhythm, melody, and sound.	Musicians, Singers, Composers	Pitch discrimination; Musical memory
5	**Bodily-Kinesthetic**	Ability to control body movements skillfully.	Athletes, Dancers, Surgeons	Coordination; Dexterity
6	**Interpersonal**	Ability to understand other people.	Teachers, Counselors, Leaders	Empathy; Social sensitivity; Communication skills
7	**Intrapersonal**	Understanding oneself.	—	Self-awareness; Emotional understanding; Insight into personal strengths and weaknesses

8	**Naturalistic**	Ability to recognize patterns in nature.	Botanists, Environmentalists, Farmers	Classification of plants and animals; Sensitivity to natural surroundings

Merits
• Broadens the concept of intelligence.
• Useful in education.
• Recognizes individual differences.

Criticisms
• Limited empirical support.
• Difficult to measure scientifically.
• Some intelligences may resemble talents rather than intelligence.

III Sternberg's Triarchic Theory of Intelligence

Robert Sternberg, 1985

Sternberg argued that traditional IQ tests measure only a part of intelligence. He suggested that intelligence involves three interacting components.

Components of the Theory

Component	Also Known As	Description	Characteristics	Examples
Analytical Intelligence	*Componential Intelligence*	Ability to analyze, evaluate, compare, and solve problems.	Logical reasoning; Academic success; Critical thinking	Solving mathematical problems; Answering examination questions
Creative Intelligence	*Experiential Intelligence*	Ability to deal with novelty and generate innovative ideas.	Imagination; Creativity; Adaptation to unfamiliar situations	Inventing new methods; Producing original artwork; Scientific discoveries
Practical Intelligence	*Contextual Intelligence*	Ability to apply knowledge effectively in everyday life.	Social competence; Decision-making; Environmental adaptation	Managing interpersonal relationships; Handling workplace challenges; Solving real-life problems

Sternberg emphasized that many successful individuals excel because they possess strong practical intelligence, even if they do not score exceptionally high on traditional IQ tests.

Merits
- Provides a broader understanding of intelligence.
- Includes creativity and practical skills.
- Has significant educational applications.

Criticisms
- Measurement of creative and practical intelligence is challenging.
- Boundaries among the three components are sometimes unclear.

3. Comparison of Theories

Aspect	Spearman	Gardner	Sternberg
View of Intelligence	Single general ability	Multiple independent intelligences	Three interacting abilities
Number of Components	Two (g and s)	Eight or more	Three
Focus	Cognitive ability	Diverse talents	Real-world adaptation
Assessment	IQ tests	Educational observation	Broader performance measures
Major Contribution	General intelligence concept	Recognition of diverse abilities	Inclusion of creativity and practicality

Conclusion

Morgan and King conceptualize intelligence as a general capacity for learning, reasoning, problem-solving, and adaptation. While Spearman emphasized a single underlying mental ability, Gardner proposed multiple relatively independent intelligences, and Sternberg highlighted analytical, creative, and practical dimensions. Together, these theories provide a comprehensive understanding of human intellectual functioning and its application in education, work, and daily life.